20. Dissolved O_2 Content A pollution control inspector suspected that a river community was releasing amounts of semitreated sewage into a river. To check his theory, he drew five randomly selected specimens of river water at a location above the town, and another five below. The dissolved oxygen readings (in parts per million) are as follows:

Above Town	4.8	5.2	5.0	4.9	5.1
Below Town	5.0	4.7	4.9	4.8	4.9

a. Do the data provide sufficient evidence to indicate that the mean oxygen content below the town is less than the mean oxygen content above? Test using $\alpha = .05$.

b. Estimate the difference in the mean dissolved oxygen contents for locations above and below the town. Use a 95% confidence interval.

a. $X_a = \{\text{Above Town}\}$

$X_b = \{\text{Below Town}\}$

$$\bar{X}_a = \frac{4.8+5.2+4.9+5.1+5.0}{5} = 5$$

$$\bar{X}_b = \frac{5.0+4.7+4.9+4.8+4.9}{5} = 4.86$$

$S_a = 0.1581 \quad S_b = 0.11401$

$\frac{S_b}{S_a} < 3 \Rightarrow$ assume $\sigma_A = \sigma_b$

$$S_p^2 = \frac{(n_a-1)S_a^2 + (n_b-1)S_b^2}{n_a+n_b-2}$$

$$= \frac{4(0.1581)^2 + 4(0.1140))^2}{8}$$

$$= 0.0189$$

$$t^* = \frac{(\bar{X}_a - \bar{X}_b) - 0}{\sqrt{\frac{S^2}{n_a} + \frac{S^2}{n_b}}} = \frac{5-4.86}{\sqrt{0.0189\left(\frac{2}{5}\right)}} = 1.610$$

$t_{8;0.025} = 2.306.$

rejection region: $-2.306 > t$, $t > 2.306$

$\Rightarrow$ not reject $\sigma_a = \sigma_b$

b.

$$CI: (\bar{X}_a - \bar{X}_b) \pm t_{8;0.025} \cdot \sqrt{S^2\left(\frac{1}{n_a} + \frac{1}{n_b}\right)}$$

$= (5-4.86) \pm 2.306 \cdot 0.0869$

$= 0.14 \pm 0.2004$

$= (-0.0604, 0.3404)$